Soil and Growing Media

Science

May, 2026

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South East Technological University

Soil and Growing Media (A08581)

Short Title:	Soil and Growing Media		
Faculty	Science and Computing		
Department	Science		
Cluster	Horticulture		
Author	GMCMAHON		
Credits	5	**Level:**	Introductory

Description of Module / Aims

This module is designed to equip students with a basic knowledge of soils including soil formation, composition and properties. Students will also learn about the macronutrients and micronutrients involved in plant nutrition. The student will learn how to carry out soil sampling and a range of soil analyses. The student will carry out an analysis of a selected soil.

Programmes

`HORT-0037`	BSc in Horticulture (`WD_SHORT_D`)	1 / 1 / M
`HORT-0037`	BSc in Horticulture (`WD_SHORT_DX`)	1 / 1 / M

Indicative Content

- Soil formation, composition and classification
- Physical, chemical and biological properties of soils and growing media
- Plant nutrition
- Soil sampling and analysis

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Describe the processes involved in the formation of soil.
2. Describe the physical, chemical and biological properties of soils and growing media and the classification of soils.
3. Discuss the role in plants and sources of macronutrients and micronutrients.
4. Discuss a range of alternative growing media.
5. Interpret a soil report and determine fertiliser requirements.
6. Use a range of laboratory techniques to characterise soils and growing media.

Learning and Teaching Methods

- Lectures
- Laboratory practical's
- Fieldwork

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,2,3,4
Continuous Assessment	50%	
Practical	40%	2,6
Assignment	10%	5

Assessment Criteria

<40%: Unable to outline key concepts. Cannot perform basic soil analysis.

40%-49%: Basic knowledge of the key learning outcomes including soil formation, composition and properties. Can perform basic soil analysis.

50%-59%: As well as the above, demonstrates an understanding of some of the complexities of soil management.

60%-69%: In addition to the above, the student demonstrates a more detailed knowledge of soil management and a high level of competence in the area.

70%-100%: All of the above to an excellent level. Advanced knowledge of the topic is supported by superior skills.

Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the final exam and continuous assessment components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	30	
Practical	18	
Independent Learning	87	

Essential Material

- Ashman, M.R. and G. Puri. _Essential Soil Science: a Clear and Concise Introduction to Soil Science_. 1st ed. Massachusetts: Wiley-Blackwell, 2008.

Supplementary Material

- Adams, C., M. Early, J. Brook and K. Bamford. _Principles of Horticulture_. 6th ed. Oxford: Routledge, 2012.
- Brady, N.C. and R.R. Weil. _The Nature and Properties of Soils_. 14th ed. U.K.: Pearson, 2013.
- Foth, H.D. _Fundamentals of Soil Science_. 8th ed. U.K.: Wiley, 1990.

Requested Resources

- Lecture Room: Loose Seated
- Room Type: Chemistry Lab